

Austin Billings

Crime Management System (README FILE)

Backbone structure

The crime prevention project is a Java-based system consisting of 4 **classes** and 3 **JFrame forms**, designed to manage officer and case records in a MySQL database.

Classes

1. Crime_management: Initiates the Main JFrame form when the program is run.
2. DatabaseConnection: Establishes a connection to the crime_management database in MySQL, using JDBC driver and credentials. This class is designed to be reusable, eliminating the need to establish a new connection in every form.
3. Officer: Contains CRUD (Create, Read, Update, Delete) methods and SQL statements to manage officer records.
4. Case: Contains CRUD methods and SQL statements to manage case records.

JFrame Forms

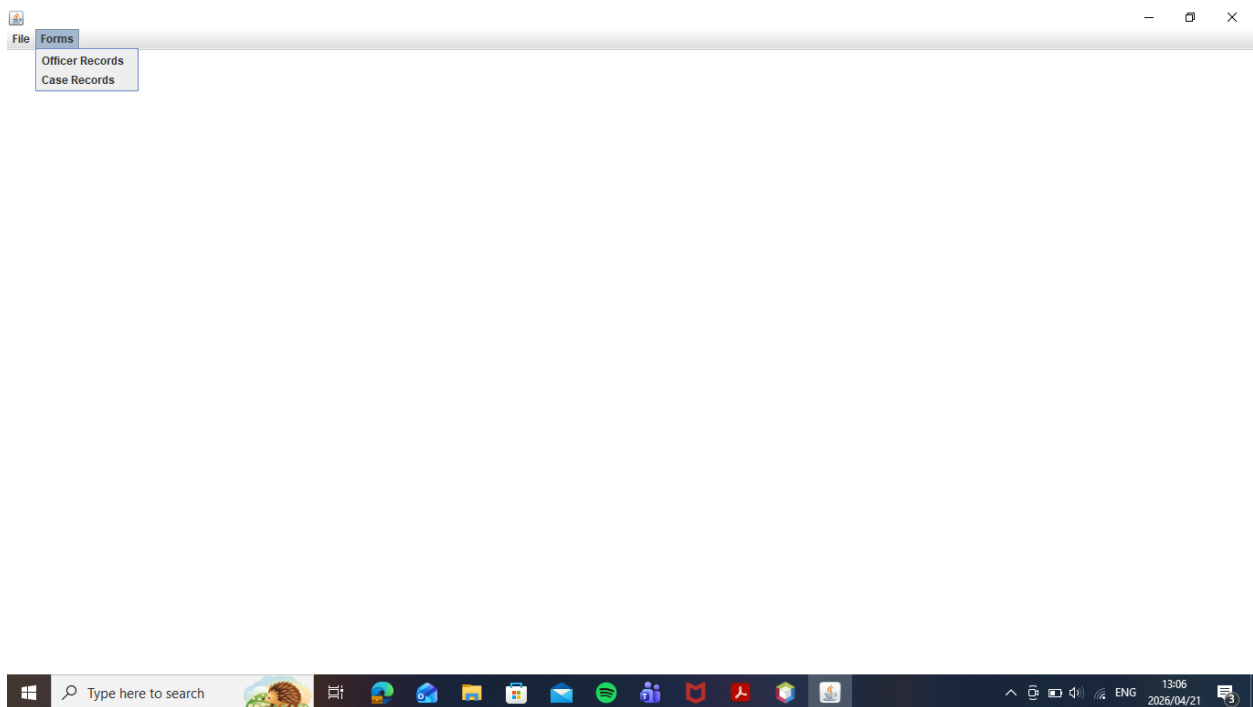
1. Main JFrame Form: A menu-driven form with options to access Officer and Case forms and exit the program.
2. Officer Form: Initiates Officer class methods, providing a user interface to interact with officer records and display data from the database.
3. The_Case Form: Initiates Case class methods, providing a user interface to interact with case records and display data from the database.

Using the System

1. Run the program, and the Main JFrame form will appear with a menu bar.
2. Click on "File" to exit the program or "Forms" to access Officer or Case records.
3. Select "Officer records" or "Case records" to access the corresponding form and interact with the database.

This design allows for a fluid user experience, with easy access to different forms and minimal complexity. By separating the database connection and CRUD operations into separate classes, the code is more organized and maintainable

Screenshots of program running in Netbeans



Officer Records Exit

First Name: Last Name:

Rank: Officer ID:

Phone:

Delete Record Update Records Add Officer

OfficerID	FirstName	LastName	Rank	Phone
5	James	Patel	Senior	10111
7	Moe	Lerato	Senior	101112
9	Bee	Dlamini	Senior	101113
10	Kai	Khan	Senior	101114
11	Bafana	Nkosi	Senior	101115

Case Records Exit

Case Details:

Case ID:

Attending Officer:

Category:

Location:

Case Status:

Reported on:

Add Case Update Cases Delete Case

CaseID	CaseDetails	Attending Officer	Category	Location	ReportedOn	CaseStatus
2	Burglary at City Mall	5	Burglary	City Mall(JHB)	2025-05-05	Under Investigation
3	Cyber Fraud Complaint	7	Fraud	Johannesburg City Pol...	2025-05-03	Open
4	Car Theft Case	9	Robbery	Johannesburg City Pol...	2025-05-01	Under Investigation
5	Assualt Report at Night...	10	Assault	Sandton Nightclub	2025-04-28	Closed
6	Domestic Dispute	11	Domestic Violence	Johannesburg City Pol...	2025-04-30	Open